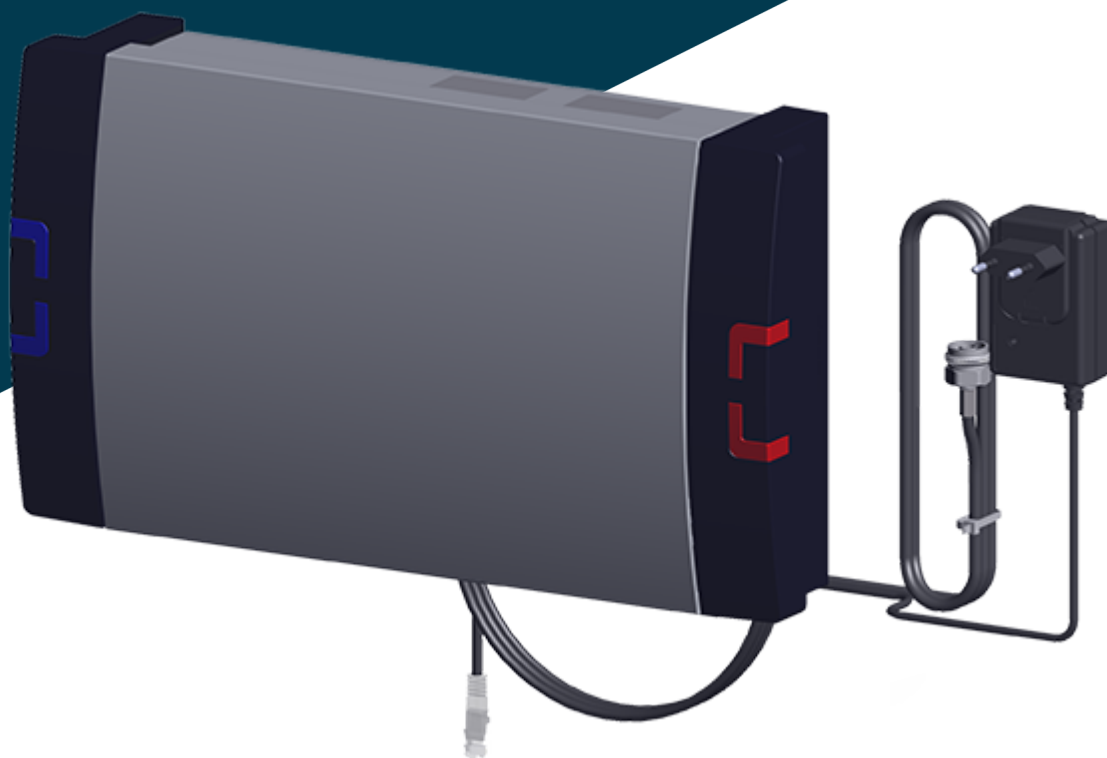


VP8002-VP4102 (VPU-UHF Reader) set



Installation manual

Version 02.000 / August 2022 / EN

Original instructions



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Warranty and spare parts

Please consult the Nedap products dealer from whom you purchased this product, in regards to the applicable warranty conditions. This product cannot be used for any other purpose as described in this document. If the product is not installed according to this document; the warranty provided is not applicable. At the sole discretion of Nedap N.V., Nedap N.V. may decide to change the conditions of the warranty policy. You agree that Nedap N.V. is able to compensate you the pro-rata value of the warranty involved rather than replacing or repairing the product depending on the technical or economical value of the product. Prior to applying the warranty, please verify if you comply with the warranty conditions of the warranty policy, whether you can successfully apply for the replacement or repair of a defective part. Parts can only be replaced with original Nedap parts, otherwise the warranty policy will not be applicable on the product. If the warranty is applicable, please contact the dealer or send the defective parts to the dealer.

Additional information

For any information or questions regarding the product, please contact your own dealer.

VP8002-VP4102 (VPU-UHF Reader) set



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1 Safety

Read this manual before using this product. Failure to follow the instructions and safety precautions in this manual may result in serious injury or death. Keep this manual in a safe location for future reference.

Symbols used in the manual

 Danger	Indicates a hazardous situation that, if not avoided, will result in death or serious injury.
 Warning	Indicates a hazardous situation that, if not avoided, could result in death or serious injury.
 Caution	Indicates a hazardous situation that, if not avoided, could result in minor or moderate injury.
	Indicates important information but not hazard related.
	Suggestions and advice to perform certain tasks more easily.

General safety instructions

-  **Warning**
Always turn off the mains power supply when working on the electrical installation.
-  **Warning**
Always wear proper protection when installing and maintaining the VP8002 and VP4102.
-  **Caution**
Installation and service should only be done by locally qualified personnel.
-  **Caution**
Install the system according to the local rules and regulations.

Working environment

-  **Caution**
The installation area must be free from any obstacles, including animals.
-  **Caution**
Make sure all components are installed out of reach of animals.



**Caution**

Make sure all cables are properly concealed, and form no danger for stumbling.

**Caution**

Take into account the high concentrations of ammonia when installing and maintaining the VP8002 and VP4102 in pig barns.

Animal welfare and safety

The automated actions of the Nedap Livestock Management systems do never discharge the installer and the user of the system from his/her responsibility to assure **and** to take care of the well-being of the animals.



2 VP8002-VP4102 (VPU-UHF Reader) set introduction

The Nedap VPU-UHF Reader set consists of a Nedap VP8002 VPU and a Nedap VP4102 UHF Reader in a V-box with power adapter, LAN/Ethernet cable and antenna cable.



Figure 1: VP8002-VP4102 set and antenna with mounting bracket (including mounting material).

 The antenna is not part of the VP8002-VP4102 set and must be ordered separately. The antenna can be ordered with an antenna mounting bracket, including mounting material. This manual describes the installation of the VP8002-VP4102 set, including the connection with the antenna and network:

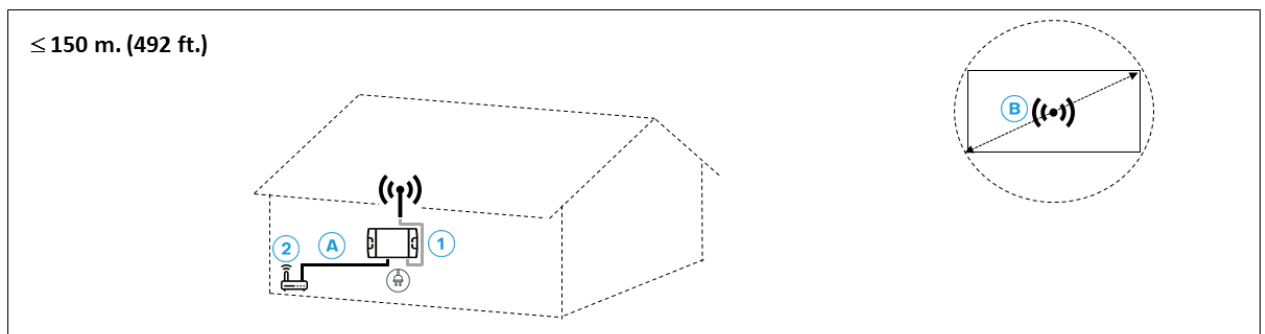


Figure 2: Example of an installation with a VP8002-VP4102 set in a dairy farm

1. VP8002-VP4102 set with antenna
2. Router

- A. LAN max. 100 m (328 ft.) or Wi-fi bridge
B. Antenna coverage 150 m (492 ft.).



3 Installation

Do not operate the product without first reading this chapter and the safety section at the beginning of this manual.



Warning

Failure to follow safety precautions in this chapter could result in serious injury or death.

3.1 Requirements for installation

3.1.1 Electrical requirements

Mains power

Make sure the mains power supply for the system is easily accessible and not too far away from the barn where the units are placed. The power sockets shall be installed near the equipment and must be easily accessible.



Caution

The VP8002-VP4102 set is provided with a double insulated power adapter. However, for future use, we advise to provide a power socket with protective earth (PE).



If there is no PE available, create a PE next to the power socket to be used for the VP8002-VP4102 set. The properties of a correct PE depend on local circumstances and legislation. Always comply with local rules and regulations when installing earth electrodes.

A power socket must be available near each VP8002-VP4102 set that meets the following requirements for the power adapter:

- Input voltage of the power adapter: 100-240 Vac
- Input current of the power adapter: 0.6-0.3A
- Input frequency: 50/60hz



Do NOT install cables directly to metal ceiling, trusses and feed lines.

Surge protection and UPS



Caution

Always use surge protectors with Ethernet surge protection.



Install an Uninterruptible Power Supply (UPS), connected to the power adapter when the power source is not reliable and constant. The UPS is intended to correctly power down the VPU (VP8002) and will also run the other V-packs in the Nedap system for a brief period. Nedap recommends a UPS size of 650 VA.



The VP8002 has internal backup power for a maximum of 30 seconds, that will correctly shut down the VP8002 when the power is interrupted. The internal backup power will not run the other V-packs in the Nedap system.



Lightning protection

It is important to follow the guidelines described in this section closely in order to minimize risk of damage on the Cow Locating system in case of lightning. Nedap does, however, not accept any responsibility for damage caused by high voltage (such as lightning), as described in the Warranty Policy.

Protective Earth (PE)

PE is meant for safety related issues such as electric shocks to humans or animals. PE will not protect devices (sufficiently) when they are struck by lightning. Normally PE is situated next to the main power source only.

Grounding

Grounding is a connection to the ground, but not Protective Earth (PE), for example a metal roof that is connected to different earth electrodes in order to lead high peak currents (such as lightning) to the earth. For the Cow Locating system all metal parts in the barn should be grounded properly.

3.1.2 Network requirements

- Router to connect the VP8002 to the Internet.
- Use of DHCP (Dynamic Host Configuration Protocol) advised.
- LAN Ethernet cable FTP minimum cat 5.
- Minimum upload speed: 1 Mb/s

When the distance between the components exceeds 100 m (328 ft), or there is a high risk of lightning strikes in the area, use optic fiber connections.



Caution

Nedap strongly advises using a local network provider. Nedap can never be kept responsible for incorrect functioning of networks or any damage arising from the recommendations that are mentioned in this document.

3.1.3 Tools requirements

- Screwdriver size 0.4 x 2.5 mm (0.016 x 0.098 in.) for the LAN connector of the V-pack.

3.2 Mount the VP8002-VP4102 set

The V-box with the VP8002 and VP4102 can be mounted either on the antenna mounting bracket or on a DIN rail that is fixed on the wall.

1. Place the VP8002-VP4102 set and antenna on the location indicated in the installation plan.



Caution

Do not install the V-box in direct sunlight.

2. Position the antenna in the barn taking the following advice into consideration:
 - a. Determine the optimum antenna position for signal reception according to the instructions in Place the antenna (page 9).
 - b. Determine the optimum mounting position of the antenna in the barn:



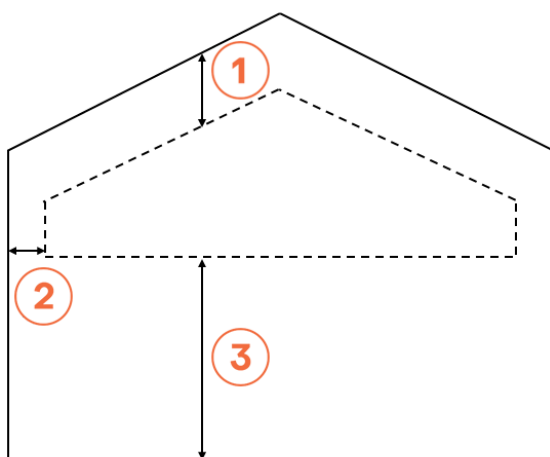


Figure 3: Antenna in barn with no ceiling

1. > 1 meter (highest point of the antenna)

2. > 0,3 meter

3. 3 - 4 meters (lowest point of the antenna. The animals should not be able to reach the antenna.)
 - - - - = area where antenna should be mounted

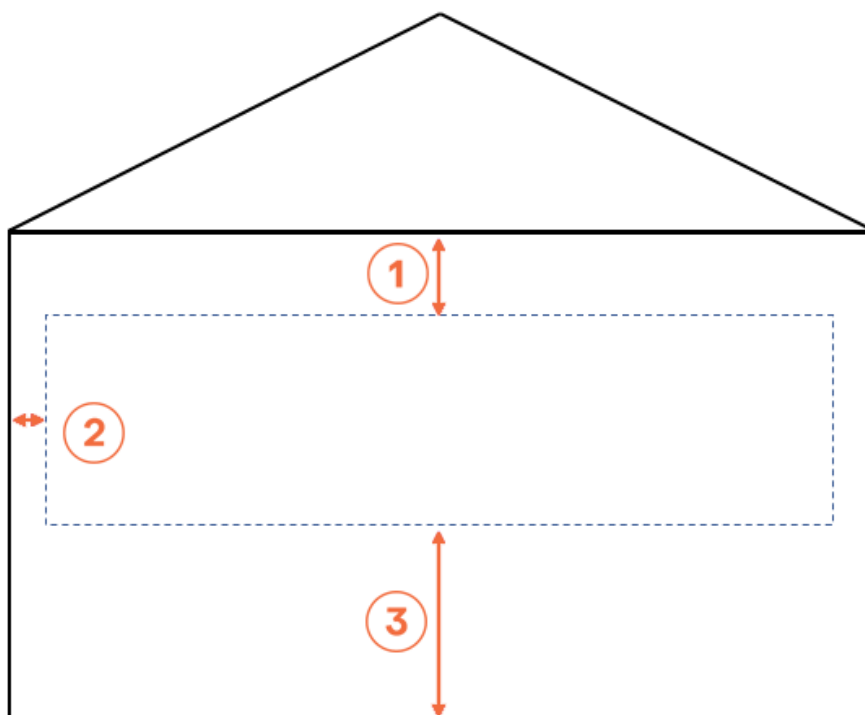


Figure 4: Antenna in barn with ceiling

1. Try to make the distance from the highest point of the antenna to the ceiling as big as possible, in order to reach an optimal signal quality.

2. > 0.3 meter.

3. In any case out of reach of the animals, preferably > 1 meter. If the antenna is not placed in a location where animals are, make sure that you can walk by comfortably.

- - - - = area where the antenna should be mounted

3. Mount the VP8002-VP4102 set:

- a. **Recommended:** Mount the set on the mounting bracket according to the instructions in the manual that is supplied with the mounting bracket. The mounting bracket can be mounted either on the wall or on a pillar of the barn.



- b. **Alternatively:** Mount the set on the rail (see Mount the VP8002-VP4102 set on the rail (page 12)).
4. Mount the antenna:
- a. Mount the antenna in the mounting bracket.
-  *When you do not use the antenna mounting bracket, follow the next guidelines for mounting the antenna:*
- *It is recommended to mount the antenna vertically (upright).*
 - *Mounting the antenna vertically upside down is not recommended due to water ingress. When it is absolutely necessary to mount the antenna vertically in upside-down position: Make sure that no water can leak into the antenna.*
 - *Make sure the antenna is installed isolated (not grounded).*
 - *Prevent the socket for the adapter against moist or accidental splashing.*
- b. Attach the antenna cable firmly to the antenna.
- c. If necessary, roll up the remaining antenna cable.
-  **Caution**
Do not shorten the antenna cable.

3.2.1 Place the antenna

Before the antenna of the VP4102 reader is mounted, it is important to determine the optimum position for the antenna to obtain the best coverage. The antenna can be placed inside or outside the barn.

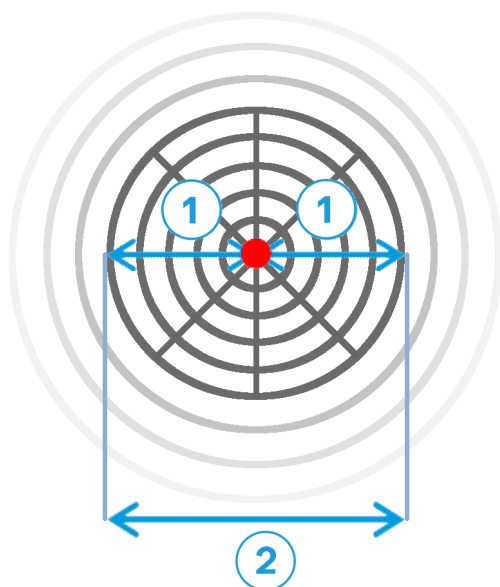
-  • *It cannot be guaranteed that the signal of the tags will pass through a (metal) wall or roof.*
- *The antenna ranges are specified for a cow activity and health monitoring system (not Cow Locating).*

 **Caution**
When the VP4102 reader is part of a Cow Locating system, follow the installation and configuration instructions in the Cow Locating Installation Manual. This manual can be obtained from your dealer.

Antenna indoor use

The range of the antenna is minimal 75 m around the antenna when placed inside the barn. Most commonly, an antenna coverage of 75 m is being achieved; a larger coverage is possible but depends on the barn layout.



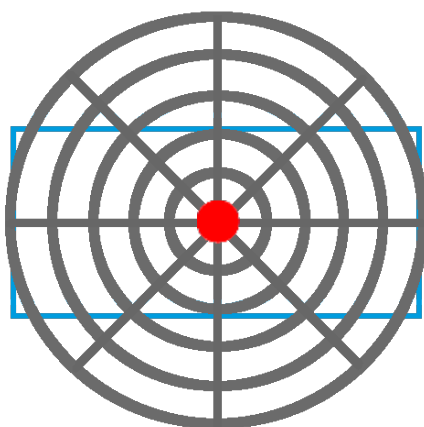


1. 0 - 75 m (0 - 82 yd)
2. 0 - 150 m (0 - 164 yd)

Red dot: Antenna

Example 1

The antenna is placed in the middle of the barn and the barn has realtime coverage.



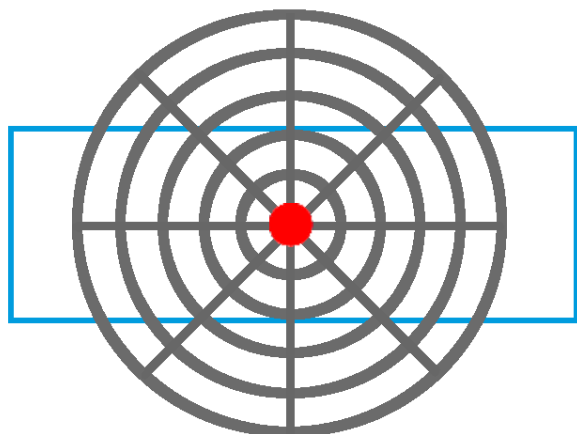
Blue rectangle: Barn

Red dot: Antenna



Example 2

The antenna is placed in the middle of the barn and the barn has no full realtime coverage. Two antennas are required to obtain full coverage (see example 3).

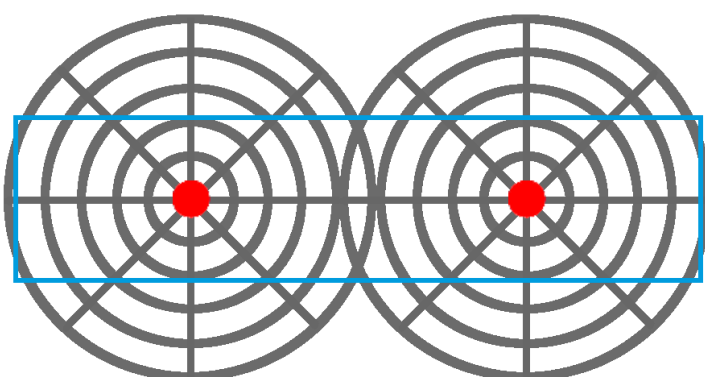


Blue rectangle: Barn

Red dot: Antenna

Example 3

Two antennas are placed so the whole barn has realtime coverage. Overlap in antenna ranges is possible to realize optimum coverage.



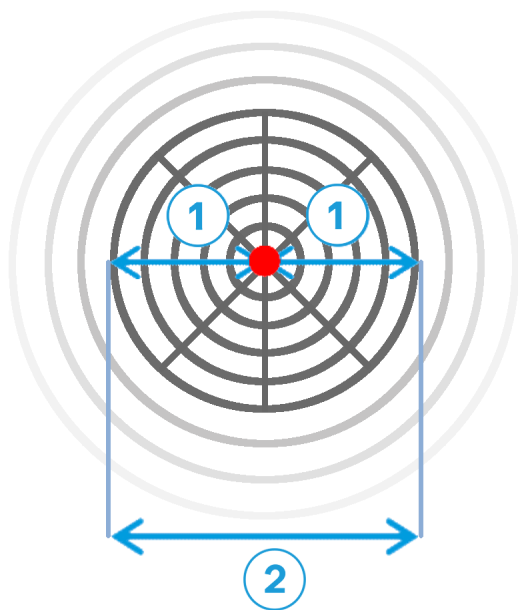
Blue rectangle: Barn

Red dot: Antenna

Antenna outdoor use

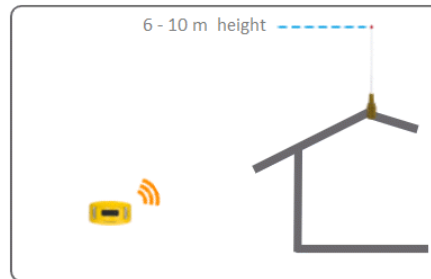
The range of the outdoor antenna is approximately 250 m (273 yd) around the antenna when placed outside the barn.





1. 0 - 250 m (0 - 273 yd)
2. 0 - 500 m (0 - 547 yd)

Red dot: Antenna

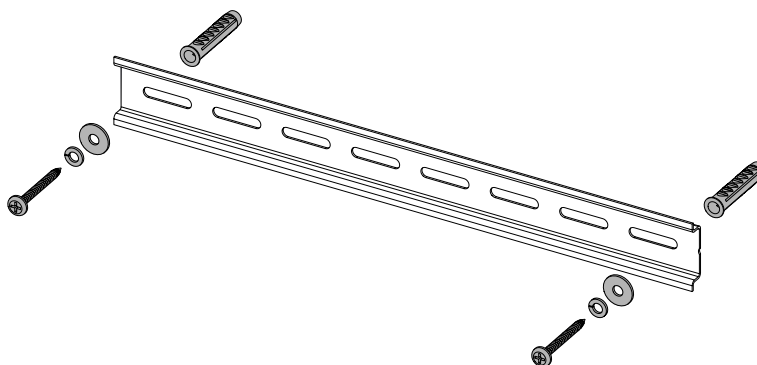


The antenna used outside only works when there is a free line of sight. The preferred height is between 8 and 10 m (26 and 31 ft) above the ground (e.g. on the roof of the barn). The 250 m (273 yd) reading range of the antenna can only be reached when there is a free line of sight (no obstacles) between the neck tag and the antenna.

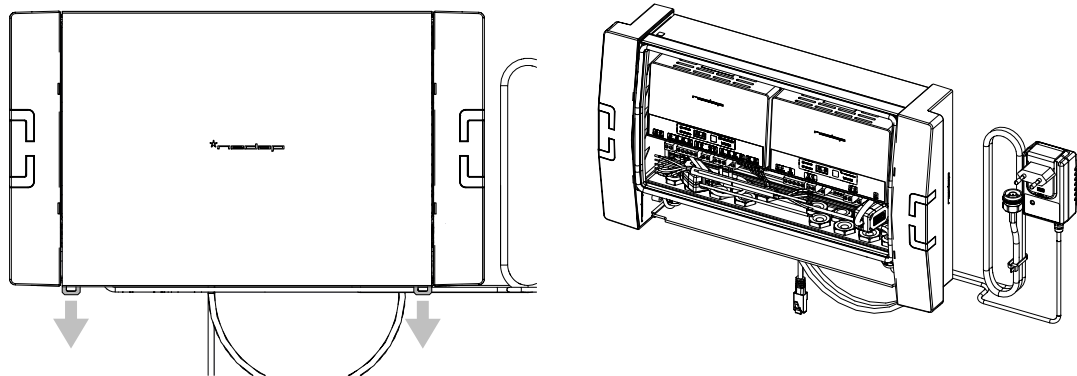
 *The long outdoor antenna has a limited signal reception close to the antenna itself. When the signal reception close to the antenna is important, the short indoor antenna can be used outside the barn.*

3.2.2 Mount the VP8002-VP4102 set on the rail

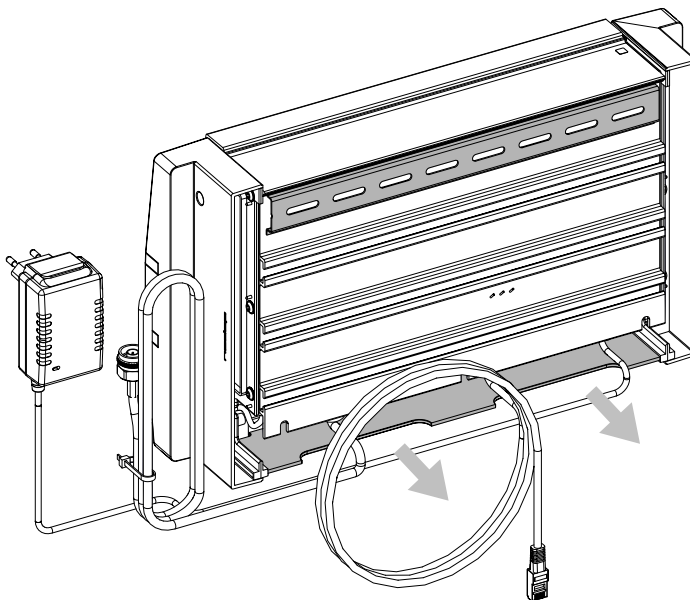
1. Mount the rail on the wall.



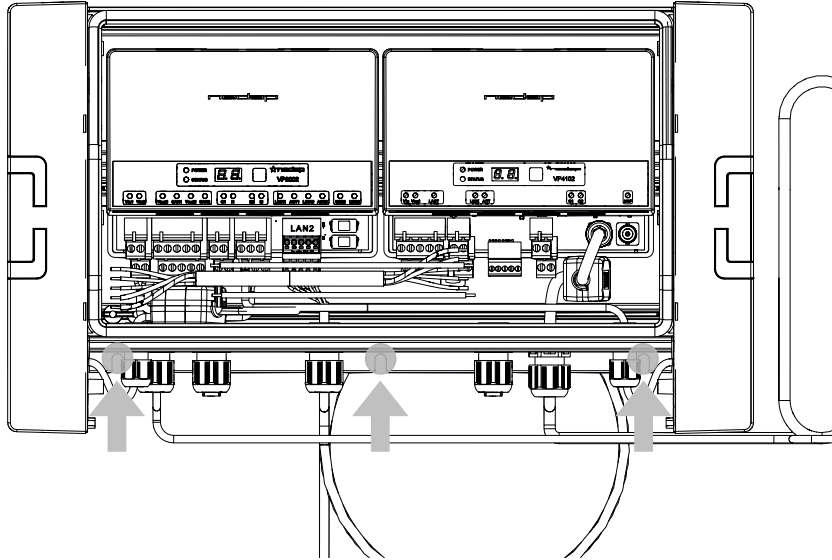
2. Open the V-box and remove the cover.



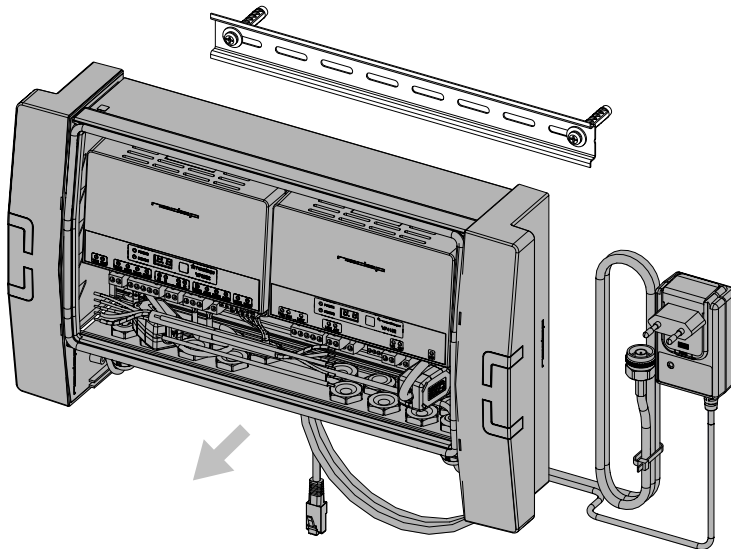
3. Put the V-box on the rail and remove the cable protection plate.



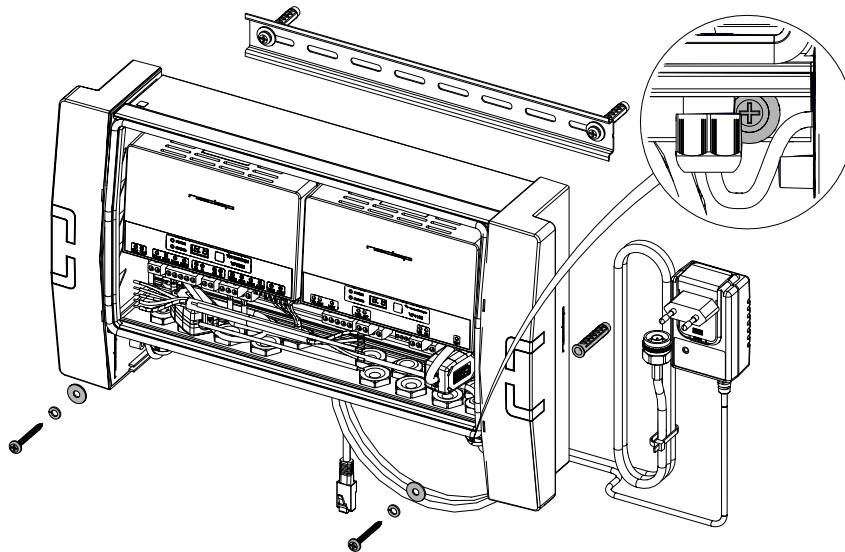
4. Mark 2 of the 3 lower points on the wall for mounting the V-box.



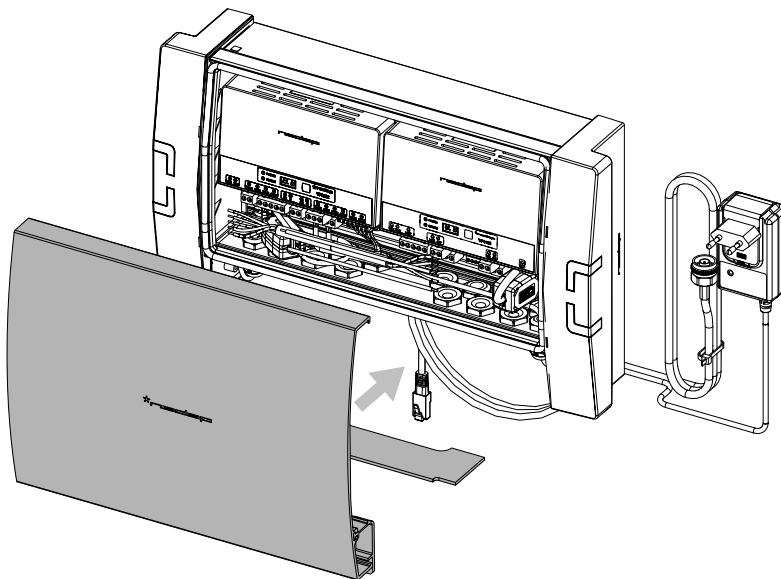
5. Remove the V-box, drill the 2 holes and place the plugs.



6. Place the V-box back on the rail and mount with 2 screws.



7. Place the cable protection plate and the cover and close the V-box.



3.3 Wiring

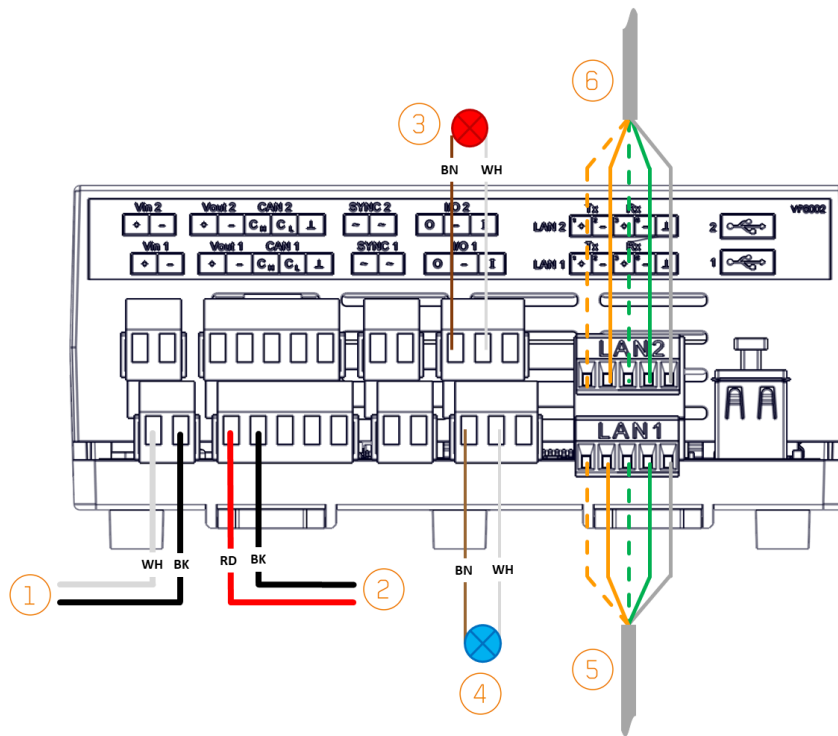
The VP8002-VP4102 set is pre-wired with a power adapter cable, LAN cable and antenna cable. For standard use, no additional wiring in the V-box needed.

1. Connect the LAN cable of the VP8002 VPU with the router.

⚠ *If the standard LAN cable needs to be replaced with a longer LAN cable (max 100 m (328 ft)), see the LAN connection in Wiring diagram VP8002.*



3.3.1 Wiring diagram VP8002 VPU



1. Power in (pre-wired power cable with adapter)
 2. Power out to VP4102 (pre-wired)
 3. V-box attention light (pre-wired)
 4. V-box status light (pre-wired)
 5. LAN 1 connection for external network (pre-wired ethernet cable)
 6. LAN 2 connection with VP4102 reader (pre-wired)
- Wiring colors: see [Wiring color coding \(page 26\)](#)

Connect the wiring of the VP8002 according to the wiring diagram:

1. Connect the pre-wired ethernet cable with the router.

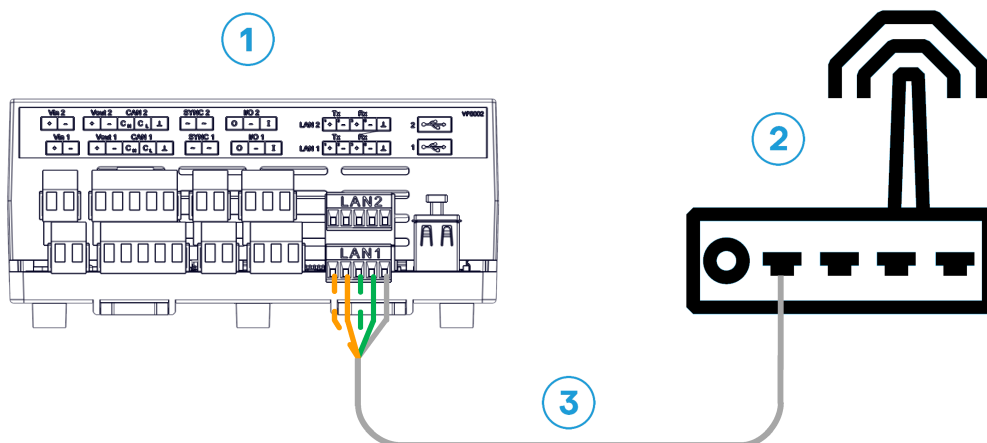


Figure 5: Connect the VP8002 with the router

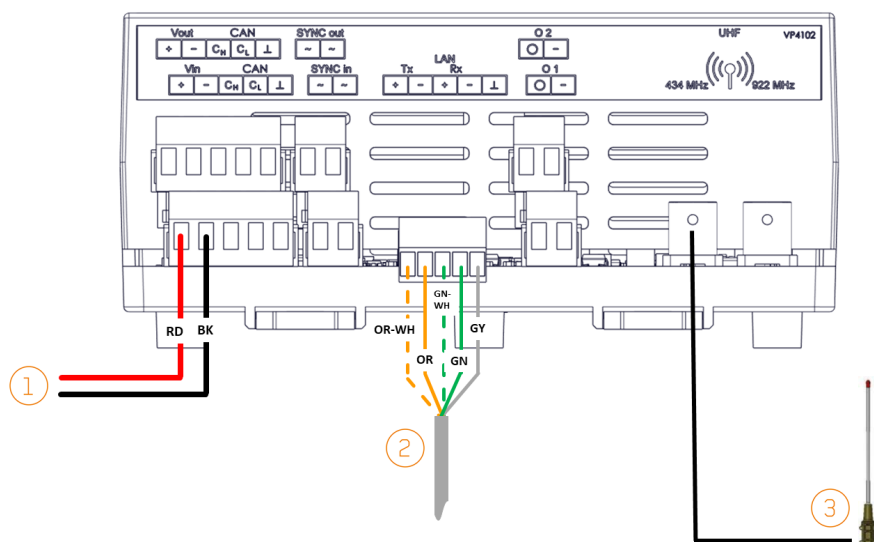
1. VP8002
2. Router

3. LAN cable



⚠ If the standard LAN cable needs to be replaced with a longer LAN cable (max 100 m (328 ft)), see the LAN1 connection in the wiring diagram.

3.3.2 Wiring diagram VP4102 reader



1. Power in from VP8002 (pre-wired)

2. LAN2 connection with VP8002 (pre-wired)

3. Antenna 434 MHz

Wiring colors: see [Wiring color coding \(page 26\)](#)

⚠ Caution

The default antenna connection is 434 MHz. For Japan and Korea switch to 922 MHz!

The VP4102 reader switches automatically to 922 MHz, if the country setting of the VPU is changed to Japan or Korea.

⚠ If the VP4102 reader is part of a complete Nedap system, use the wiring diagram in the installation manual of this system. The installation manual can be obtained from your dealer or on our Business portal: <http://www.nedap.com/livestockmanagement-portal>.



4 Configuration



Caution

When the VP8002-VP4102 set is part of a complete Nedap system (e.g. Nedap CowControl system), follow the configuration instructions in the installation manual of this product. This manual can be obtained from your dealer or on our Business portal: <http://www.nedap.com/livestockmanagement-portal>.

4.1 Power up the VP8002-VP4102 set



Caution

- Before powering up and configuring the VP8002 VPU and VP4102 reader, the system must be fully mounted and all cabling must be installed and checked.
- Contact a network specialist for a proper network configuration.

1. Plug the power adapter of the VP8002-VP4102 set in the socket to power up.
The red light on the V-box blinks 1 time. The blue light blinks every 5 seconds. When the blue light is on (breathing), the VP8002-VP4102 set is ready for use.

4.2 Find the IP address of the VP8002

If the VPU is connected to a router with DHCP (recommended), it automatically gets a unique IP address from the router it is connected to.

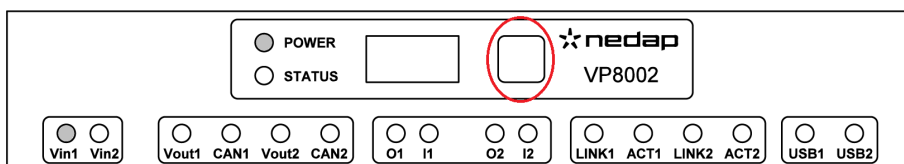
Find the IP address with the online tool Find my VPU

1. In your Internet browser, type nedap-bi.com/find_my_vpu
2. All VP8002s of the network are displayed. It can be just one, or multiple.
3. Click on the IP address of the VP8002 of your installation.
The login window for Nedap Velos appears.
4. Continue with Login to Velos and complete the service checklist (page 19)

Find the IP address manually

If the online tool does not work, you can open the V-box and use the button on the display of the VP8002:

1. Briefly push the button on the VP8002: "IP" appears in the display, followed by the IP address.



2. Write down the numbers shown in the display, for example 192.168.1.41.



- The "." is shown in the display as "-".
- You can verify the address by briefly pushing the button again.

3. Take a device (PC, tablet or smartphone) that is connected with the same network, open the Internet browser and type the IP address.
The login window for Nedap Velos appears.
4. Continue with Login to Velos and complete the service checklist (page 19)





The first time the VP8002 is connected, the Velos setup wizard will start.

IP address not found

If the IP address cannot be found, the display shows one of the following codes.

Display code	Situation	Solution
0.0.0.0	No network connection	- Check if the VP8002 and PC are connected to the same network. - Check the network or contact a network specialist.
169.254.XX	The VP8002 did not receive an IP address from the router.	

4.3 Login to Velos and complete the service checklist

After completing the setup wizard, the login window to Nedap Velos appears. Login to configure the basic settings for the system.

- Login with:
 - Username:** service
 - Password:** service (or own created password)
 After logging in, the Velos Dashboard with the service checklist appears.
- Complete the service checklist on the Velos Dashboard:
 - Set all items in the **Service** section that **do not** have a green checkmark:

4.4 Firmware updates

If a firmware update is available, a notification appears on the Dashboard:

- In Velos go to **Dashboard > System notifications > V-pack firmware**.
V-pack firmware updates are listed, if available.



The VP8002 must be updated first. Next the VP4102 reader can be updated.

- VP8002:**
 - Select the VP8002 firmware update.
 - Click on **Actions > Update**.
The VP8002 firmware is updated.
- VP4102:**
 - Select the VP4102 firmware update.
 - Click on **Actions > Update**.
The VP4102 firmware is updated.

4.5 VP4102 reader check

When the VP4102 reader is linked to the LAN network of the farm, the functioning of the VP4102 reader can be checked with the following additional tests:

- VP4102 reader check (page 27)



2. VP4102 reader analysis (page 27)
3. VP4102 tags analysis (page 28)



5 Handling Instructions

Storage

- If the product is to be stored for some time, make sure that it is under a protective cover to prevent dirt and moisture from entering.
- Do not expose the product to direct sunlight and / or adverse weather conditions such as storm, rain, hail or snow.
- See Summary technical specifications for the storage temperature range and relative humidity.

Disposal

The owner or last user of the product is responsible for the proper disposal of (parts of) the product according to local rules and regulations.

The WEEE  symbol in Europe indicates that the relevant electrical product or battery should not be disposed of as general household waste in Europe. To ensure the correct waste treatment of the product and battery, please dispose them in accordance to any applicable local laws of requirement for disposal of electrical equipment or batteries. In so doing, you will help to conserve natural resources and improve standards of environmental protection in treatment and disposal of electrical waste (Waste Electrical and Electronic Equipment Directive WEEE 2012/19/EU).



6 Technical specifications

VP8002

Item	Specification
Input voltage	25 Vdc, $\pm 10\%$. - Protected against reverse power supply connection.
Power consumption	Max. 300 mA (excluding external loads).
Output voltage	25 Vdc, $\pm 10\%$.
Dimensions (L x W x H)	Without packaging: 143 x 127 x 68 mm (5.63 x 5.0 x 2.68 in.) With packaging: 173 x 138 x 70 (6.81 x 5.43 x 2.76 in.)
Weight	Without packaging: approx. 400 g (0.88 lb) With packaging: approx. 500 g (1.10 lb)
Outputs	2 provided, max. 250 mA per output @ 25 Vdc. 2x 4 A using the CAN bus.
Inputs	2 provided. Internal pull up resistor to Vin. Input voltage < 6V: active Input voltage floating: inactive
Communication	- CAN-bus 125 kbit/s, 2 channels. - Ethernet LAN 100/10 Mbps, 2 channels (LAN1 for network, LAN2 for devices (Ethernet V-packs))
Cable specification	Min. CAT5 shielded Ethernet cable
Operating temperature range	-10 to 50 °C (14 to 122 °F); mounted in a V-box.
Storage temperature range	-20 to 70 °C (-4 to 158 °F)
Relative humidity (non-condensing)	< 93%
Enclosure protection class	- IP 20. - IP65 in V-box (if cover and cables are correctly installed).

VP4102

Item	Specification
Power supply input voltage	12 V DC - 30 V DC
Power consumption	160mA 85mA (25VDC 100 mA) (without connected I/O e.g. light)
Dimensions (L x W x H)	143 x 120 x 68 mm (excluding mounting rail)
Weight	± 279 gr
Operating temperature range	-10 to 50 °C (14 to 122 °F)
Enclosure protection class	IP30. When installed in V-box IP65 (cover and cables installed correctly!)
Relative humidity (non-condensing)	< 93%
Communication	Ethernet CAT 5e 100mbit/s max 100 m (328 ft)
Cable specification	UTP CAT5 (shielded cable is advised)



Item	Specification
Firmware	Downloadable
Antennas	G-CXL 70-1LW/h
Outputs	Max. 0.4 Amp by current limiter, short-circuiting and thermal protected
Certifications	CE, FCC, IC and KCC

Power adapter

Item	Value
Input voltage	100-240 Vac, 0.6-0.3A 50/60hz
Output voltage/current	24Vdc, 1A (UL 1310 NEC Class 2)
Operating temperature range	-20°C - 40°C (68 °F - 104 °F), indoor use
Hold up time	10 milliseconds after losing input power
Over load protection	Effected at 160% Max. load and/or +38VDC. Automatically recovering after overload being removed.
RoHS	RoHS compliant
MTBF	>150,000 hours at full load and 25°C (77 °F) ambient temperature
Certifications	CE, CSAus, CSA, EAC, KCC, PSE, CCC



7 Compliance

FCC/IC (VP8002)

This equipment has been tested and found to comply with the limits for a class B digital devices, pursuant to part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequent energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does not cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and receiver.
- Connect the equipment into an outlet on a circuit different from that to which the receiver.
- Consult the dealer or an experienced radio/TV technician for help

This Class B digital apparatus complies with Canadian ICES-003. Cet appareil numérique de Classe B est conforme à la norme Canadienne ICES-003.

Any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Les changements ou modifications n'ayant pas été expressément approuvés par la partie responsable de la conformité peuvent faire perdre à l'utilisateur l'autorisation de faire fonctionner le matériel.

FCC and ISED Compliance statement (VP4102)

This device complies with part 15 of the FCC Rules and to RSS-210 of Innovation, Science and Economic Development Canada. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

Changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate the equipment.

Cet appareil se conforme aux normes CNR 210 exemptés de licence d'Innovation, Sciences et Développement économique Canada. L'opération est soumise aux deux conditions suivantes: (1) cet appareil ne doit causer aucune interférence, et (2) cet appareil doit accepter n'importe quelle interférence, y compris l'interférence qui peut causer une opération non souhaitée de cet appareil.

Les changements ou modifications n'ayant pas été expressément approuvés par la partie responsable de la conformité peuvent faire perdre à l'utilisateur l'autorisation de faire fonctionner le matériel.

CE - UKCA declaration

Hereby NEDAP N.V. declares that the subject equipment is in compliance with:

- For CE: Directives 2014/30/EU (Electromagnetic compatibility), 2014/53/EU (Radio Equipment Directive) and 2011/65/EU (Restriction of the use of certain hazardous substances).
- For UKCA: SI 2016/1091 (Electromagnetic Compatibility Regulations 2016), SI2017/1206 (UK Radio Equipment Regulations 2017) and SI2012/3032 (Restriction of the Use of Certain Hazardous Substances in Electrical and Electronic Equipment Regulations 2012 (RoHS)).

The full text of the declarations of conformity is available at <http://www.nedap.com/livestockmanagement-portal> where, if applicable, also REACH information can be found.



Mexico

Descripción: CONTROLADOR DE IDENTIFICACION POR RADIOFRECUENCIA

Especificaciones Eléctricas: 25V $\equiv$ 1A

Marca: Nedap



8 Appendix

8.1 Wiring color coding

Table 1: Wiring colors

Color code	Wire color
BK	Black
BL	Blue
BN	Brown
GN	Green
GY	Gray
OR	Orange
PK	Pink
RD	Red
VT	Violet
WH	White
YW	Yellow
BK-WH	Black-White
BL-WH	Blue-White
BN-WH	Brown-White
GN-WH	Green-White
GY-PK	Gray-Pink
OR-WH	Orange-White
RD-BL	Red-Blue
YW-GN	Yellow-Green
YW-WH	Yellow-White

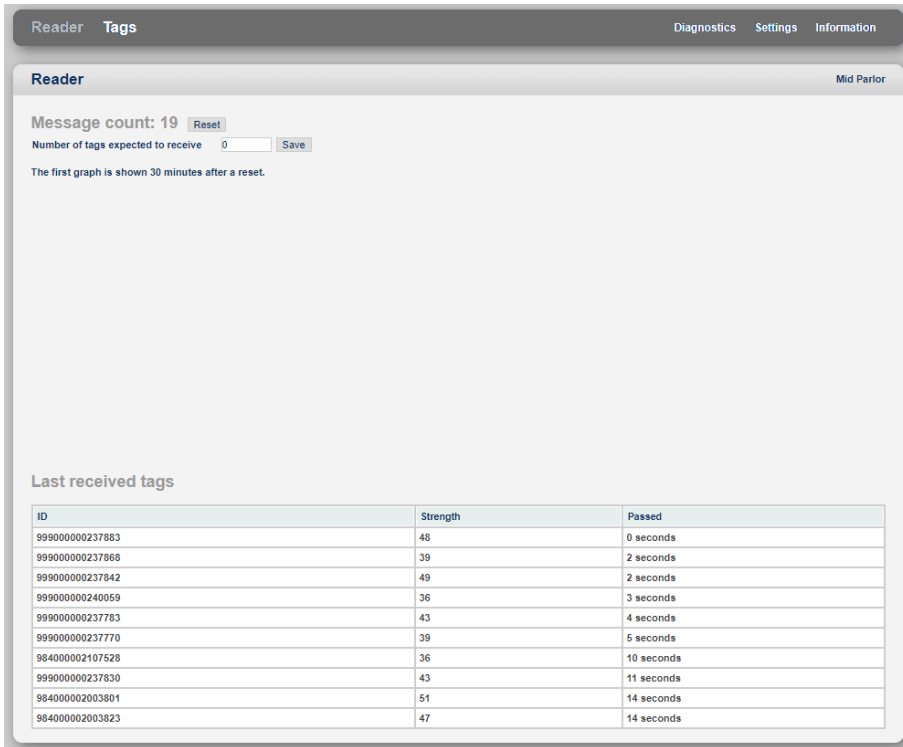


8.2 VP4102 reader check

When the VP4102 reader is linked to the LAN network of the farm, the functioning of the reader can be checked in the **Reader** window. The diagram in this window shows if the VP4102 reader is receiving the data of the tags properly.

 The blue light on the V-box blinks each time a message from a tag is received by the VP4102 reader.

1. Go the **Reader** tab of the VP4102 reader web page.



ID	Strength	Passed
999000000237883	48	0 seconds
999000000237868	39	2 seconds
999000000237842	49	2 seconds
999000000240059	36	3 seconds
999000000237783	43	4 seconds
999000000237770	39	5 seconds
984000002107528	36	10 seconds
999000000237830	43	11 seconds
984000002003801	51	14 seconds
984000002003823	47	14 seconds

2. The IP address of the VP4102 reader is displayed in the upper right corner of the window. For ease of use, you can replace this IP address with a name to indicate the location in the barn:
 - a. Click on the IP address: the name field opens in the upper right corner
 - b. Type the name of the location in the barn where the VP4102 reader is mounted.
 - c. Click **Save**.
3. The diagram in the **Reader** page shows the number of messages sent to the VP4102 reader during the last 24 hours. The following information can be set or read:
 - **Message count:** The total number of messages (received since start-up) about activities like rumination and eating.
 - **Number of tags expected to receive:** 0 at start-up.
 - **Last received tags:** Lists the tags that sent the last received messages, including the tag number, signal strength and time passed since the message was received.

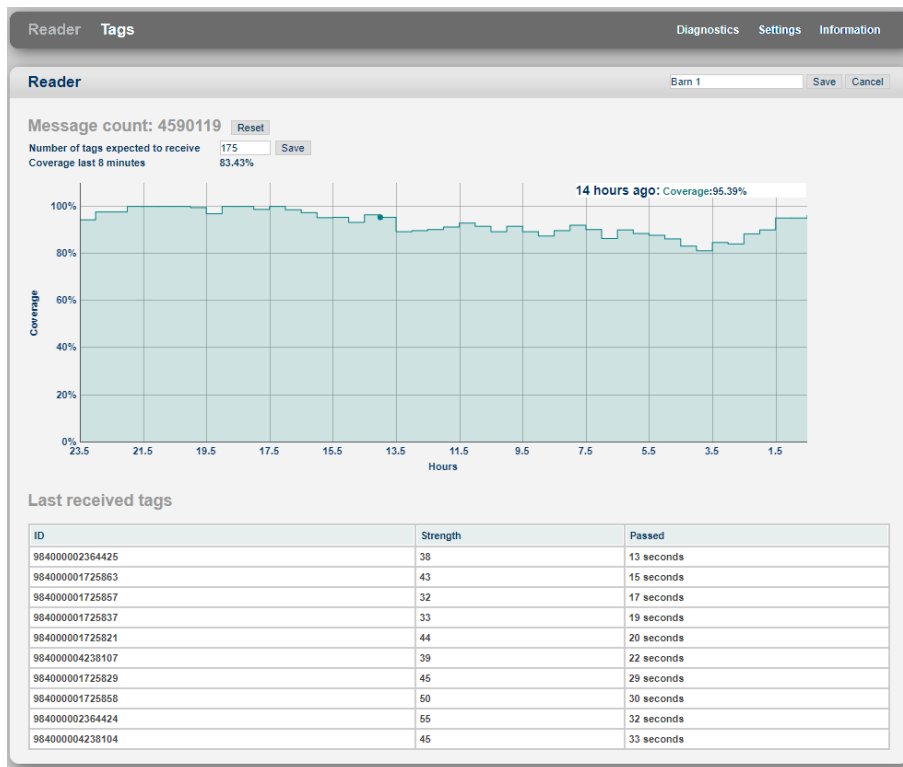
 The signal strength ranges from 1 (weak) to 80 (very strong).

8.3 VP4102 reader analysis

The operation of the VP4102 reader can be analysed when it has collected data over several hours:



1. Go the **Reader** tab of the VP4102 reader web page.



2. The diagram in the **Reader** page shows the percentage of tags that have sent messages to the VP4102 reader during the last 24 hours. The following information can be set or read:
 - **Message count:** The total number of messages (received since start-up) about the cows activities like rumination and eating.
 - **Number of tags expected to receive:** The total number of tags for this reader.
 - ! When the number is higher than 0, the y-axis annotation changes from **Message count** to **Coverage (%)**.
 - **Coverage last 8 minutes:** An indication of the percentage of tags that have sent messages during the last 8 minutes.
 - **X hours ago:** When the mouse pointer is placed in the diagram, an indication is given of the percentage of tags that have sent messages during the past X hours.
 - **Last received tags:** Lists the tags that sent the last received messages, including the tag number, signal strength and time passed since the message was received.
 - ! The signal strength ranges from 1 (weak) to 80 (very strong).
 - ! The graph is an indication of the reception and not exactly the reality. A mix of (I)FE and (I)FER(P) tags cannot be calculated correctly. Each type of message is sent with a 1 day history. Therefore not all messages need to be received for a good functioning.

8.4 VP4102 tags analysis

The reception of the tags can be monitored in the **Tags** window:



Reader
Tags
Diagnostics
Settings
Information

Tags
Mid Parlor

Add tags automatically

Automatically add tags to the list when their strength is above the threshold.

Auto add enabled: ☐

Minimum strength:

Save

Add tags manually

Enter the last eight digits of the responder number.

Add

Tag list

When one or more of the flash light boxes is checked, output 2 of the reader will only flash when receiving messages from these tags.

ID	Last	Average	Minimum	Maximum	Number Received	Last message	Flash light	Trace CPS	Remove
237764	61	Clear	Clear	Clear	Clear	29-10-2019 15:23:29	☐	☐	✗
237784	53	54	47	63	7	29-10-2019 15:23:11	☐	☐	✗
237788	48	52	43	60	7	29-10-2019 15:24:41	☐	☐	✗
237790	51	52	47	54	5	29-10-2019 15:21:42	☐	☐	✗
237821	54	49	43	54	6	29-10-2019 15:23:37	☐	☐	✗
237844	41	48	41	55	4	29-10-2019 15:16:01	☐	☐	✗
237879	36	48	35	57	7	29-10-2019 15:23:36	☐	☐	✗

Reset All Tags

- You can automatically or manually add tags that need to be monitored.
- The **Tag list** shows per tag the last 8 digits of the tag number and strength of the received messages (ranging from 1 (weak) to 80 (very strong))
- Flash light:** When this option is selected for one or more tags, the blue light of the V-box will only blink when messages from the selected tags are received.



Caution

Do not forget to deselect this option if the V-box must blink again for all received messages.

- Trace CPS:** When this option is selected for (I)FERP tags, the Cow Locating messages are also counted. This means that the message count increases faster. This makes Cow Locating (CPS) tags very suitable to test the range of reception due to the higher rate of messages.



Caution

Be aware that this test is a global approach of the reception range.



The **Tags** window can be used as a reception test:

1. Add a (I)FERP tag manually.
2. Select the option **Trace CPS**.
3. Take the tag and walk away from the antenna.
4. Use the **Tag list** to check the strength of the received messages when the distance from the antenna increases.





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